

Sri Shanmugha College of Engineering and Technology
(An Autonomous Institution)
Department of Artificial Intelligence and Data Science
CIA - I Assignment

Academic Year: 2025-2026

Year/Sem: III / V

Batch: 2023 - 2027

S.No	Roll No.	SID	Name	Assignment Questions
1	732724321001	E23AI021	Aatheeswaran B	Theory: What is the significance of Exploratory Data Analysis (EDA) in data science?
				Chart: Draw a simple line plot showing sales of a shop for 7 days.
				Case Study: A student collected marks of 5 friends in mathematics. How can you explore the data to check who scored the
2	732724321002	E23AI050	Abu shefin EC	Theory: Differentiate between EDA and classical analysis.
				Chart: Create a histogram of ages of 10 people.
				Case Study: A teacher wants to know the average performance of students in an exam. Suggest a simple EDA method to
3	732724321003	E23AI001	Adithyan S	Theory: Why do we need data transformation in EDA?
				Chart: Plot a scatter plot showing the relation between hours studied and marks scored by 8 students.
				Case Study: A shopkeeper records the number of customers in the morning and evening. How can EDA help him
4	732724321004	E23AI020	Dharunkumar S	Theory: What is the role of visual aids in EDA?
				Chart: Draw a bar chart showing sales of four products: A, B, C, and D.
				Case Study: A bakery records sales of cakes on weekdays and weekends. How can visualization help in understanding
5	732724321005	E23AI037	Girimurugan A	Theory: Write a short note on pivot tables in EDA.
				Chart: Create a density plot for heights of 6 students.
				Case Study: A teacher maintains a database of students with subjects and marks. How can pivot tables help summarize the
6	732724321006	E23AI047	Gokul K	Theory: Explain the use of grouping datasets in EDA.
				Chart: Plot a pie chart showing the percentage of students choosing Science, Commerce, and Arts.
				Case Study: A college wants to know the distribution of students across different departments. How can grouping help?
7	732724321007	E23AI017	Gowtham G	Theory: What are cross-tabulations in EDA?
				Chart: Plot a line chart showing temperature variation during morning, afternoon, and night for 5 days.
				Case Study: A mobile store wants to know whether young or older customers buy more smartphones. How can cross-
8	732724321008	E23AI070	Gowtham P	Theory: What are the advantages of using Matplotlib for visualization?
				Chart: Create a scatter plot comparing height and weight of 6 students.
				Case Study: A doctor records the age and weight of patients. How can a scatter plot help him analyze the relation?
9	732724321009	E23AI025	Hari E	Theory: Explain the concept of subplots in Matplotlib.
				Chart: Draw two line plots in a single figure: one showing monthly income and another showing monthly expenses.
				Case Study: A family wants to compare monthly savings with monthly expenses. How can subplots help in visualization?
10	732724321010	E23AI063	Harish Pirajin V	Theory: What is the purpose of legends in plots?
				Chart: Plot a line graph showing marks of three students in five subjects with legends.

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				Case Study: A school wants to compare the marks of three students in different subjects. How can legends help in the
11	732724321011	E23AI054	Himanshu ranjan S	Theory: What is the importance of customization in Matplotlib?
				Chart: Plot a bar chart showing the number of books sold in Jan, Feb, Mar, and Apr.
				Case Study: A library records book borrowing for 4 months. How can a customized chart help in identifying the most
12	732724321012	E23AI027	Mohammed shaham T	Theory: Explain the use of annotations in a plot.
				Chart: Draw a line plot showing the population growth of a town for 5 years with an annotation on the highest value.
				Case Study: A city records population growth. How can annotations highlight the year with maximum growth?
13	732724321013	E23AI058	Moushik R	Theory: Define histogram and its use in data visualization.
				Chart: Plot a histogram showing the marks of 8 students in a test.
				Case Study: A teacher wants to check how many students fall in different marks ranges. How can a histogram help?
14	732724321014	E23AI053	Muhammed sanah KP	Theory: What are density and contour plots?
				Chart: Create a contour plot for some temperature values across two regions.
				Case Study: A weather department records temperatures in different zones. How can contour plots help them?
15	732724321015	E23AI029	Prasanna S	Theory: What is the role of colors in Matplotlib visualizations?
				Chart: Plot a bar chart with different colors showing sales of 5 fruits.
				Case Study: A fruit seller wants to display sales in a colorful way to make comparison easier. How can colored bars help?
16	732724321016	E23AI033	Pravin K	Theory: What is three-dimensional plotting in Matplotlib?
				Chart: Draw a simple 3D scatter plot showing x, y, z values for 5 points.
				Case Study: An engineer wants to display 3D coordinates of machine parts. How can 3D scatter plots help him?
17	732724321017	E23AI074	Ragul B	Theory: Write short notes on visualization with Seaborn.
				Chart: Create a Seaborn barplot showing marks of students in 3 subjects.
				Case Study: A teacher wants a cleaner chart style for marks visualization. How can Seaborn help in making it attractive?
18	732724321018	E23AI041	Saravana kumar KM	Theory: Explain the use of Basemap in Matplotlib.
				Chart: Plot a simple geographic map showing three cities marked.
				Case Study: A travel company wants to highlight three tourist places on a map. How can Basemap help them?
19	732724321019	E23AI019	Seenivasan G	Theory: What is univariate analysis?
				Chart: Plot a histogram of the salaries of 6 employees.
				Case Study: A company wants to check the spread of salaries of its employees. How can univariate analysis help?
20	732724321020	E23AI051	Shahil mohammed	Theory: Define distribution in the context of univariate analysis.
				Chart: Draw a line plot showing the distribution of test scores of 10 students.
				Case Study: A tutor wants to see how test scores are spread out in a class. How can distributions help?
21	732724321021	E23AI039	Sridharan V	Theory: What are numerical summaries in univariate analysis?
				Chart: Create a boxplot for the marks of 8 students.
				Case Study: A teacher wants to quickly identify median marks and outliers. How can a boxplot help?
22	732724321022	E23AI057	Srinivas P	Theory: Define scaling in univariate analysis.
				Chart: Draw a line chart showing scaled values of marks (out of 1) for 5 students.

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				Case Study: A tutor wants to compare marks of two tests scored out of different totals. How can scaling help?
23	732724321023	E23AI015	Surendar P	Theory: What is standardization and why is it useful?
				Chart: Create a scatter plot of standardized height and weight values of 6 people.
				Case Study: A fitness trainer wants to compare height and weight on the same scale. How can standardization help?
24	732724321024	E23AI068	Suresh k	Theory: Explain inequality in the context of univariate analysis.
				Chart: Draw a bar plot showing income of 5 families.
				Case Study: A researcher wants to see if all families earn equally or if inequality exists. How can a bar chart show this?
25	732724321025	E23AI013	Suresh kumar G	Theory: What is smoothing in time series?
				Chart: Plot a line graph of stock prices over 7 days and smooth the line.
				Case Study: An investor wants to remove small fluctuations and see overall stock trends. How can smoothing help?
26	732724321026	E23AI028	Sutheeshwaran B	Theory: Why is EDA considered the first step in data science?
				Chart: Draw a histogram showing the number of cars passing a traffic signal in 6 hours.
				Case Study: A traffic police officer wants to analyze which hour has the most traffic. How can EDA help?
27	732724321027	E23AI064	Thavin GP	Theory: How does Bayesian analysis differ from EDA?
				Chart: Create a scatter plot comparing study hours vs marks for 7 students.
				Case Study: A student wants to know if more study hours always give higher marks. How can EDA support this?
28	732724321028	E23AI023	Thirumalai selvan J	Theory: What are the advantages of reshaping data?
				Chart: Draw a pivot table showing sales of 2 products across 3 months.
				Case Study: A shopkeeper wants to compare monthly sales of two products. How can reshaping data help him?
29	732724321029	E23AI045	Vishal M	Theory: What is data aggregation?
				Chart: Create a bar chart showing total sales in Q1, Q2, Q3, and Q4.
				Case Study: A company wants to know total sales for each quarter. How can aggregation be applied here?
30	732724322001	E23AI024	Abijitha DS	Theory: Why are software tools like Python important for EDA?
				Chart: Draw a line plot of website visitors for 5 days.
				Case Study: A blogger wants to analyze how many people visited his website daily. How can Python help in EDA?
31	732724322002	E23AI048	Abinaya K	Theory: What does “making sense of data” mean in EDA?
				Chart: Draw a scatter plot showing age vs monthly expenses of 6 people.
				Case Study: A family records age and expenses. How can EDA help understand spending patterns?
32	732724322003	E23AI002	Aishwarya S	Theory: Write a short note on reshaping data.
				Chart: Create a pivot table showing attendance of students in 3 subjects.
				Case Study: A teacher wants to see how attendance varies across subjects. How can reshaping data help?
33	732724322004	E23AI016	Ammu R	Theory: What is the purpose of visualizing errors in Matplotlib?
				Chart: Plot a line chart with error bars for sales of a product in 5 months.
				Case Study: A company records sales but with small variations. How can error bars show uncertainties?
34	732724322005	E23AI066	Arthi S	Theory: Why are legends important in scatter plots?
				Chart: Create a scatter plot comparing marks of boys and girls in a subject with different colors.

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				Case Study: A school wants to compare male and female student performance. How can legends help?
35	732724322006	E23AI005	Devi K	Theory: What are annotations used for in plots?
				Chart: Draw a line plot of temperature across 6 days and annotate the highest temperature.
				Case Study: A weather station wants to highlight the hottest day of the week. How can annotation help?
36	732724322007	E23AI046	Dharani sri RK	Theory: Define contour plot.
				Chart: Plot a contour graph of rainfall across two regions.
				Case Study: A farmer wants to know how rainfall varies across two farmlands. How can contour plots help?
37	732724322008	E23AI012	Dharshana G	Theory: Why do we use histograms instead of bar charts in some cases?
				Chart: Draw a histogram of exam scores of 12 students.
				Case Study: A teacher wants to see the frequency distribution of student marks. How can histograms help?
38	732724322009	E23AI075	Dheeksha P.M	Theory: What is the role of subplots in data comparison?
				Chart: Create two subplots: one showing student marks in Maths, another in English.
				Case Study: A tutor wants to compare subject-wise performance. How can subplots make it easier?
39	732724322010	E23AI030	Durga G	Theory: Explain “density plots” in data visualization.
				Chart: Draw a density plot of salaries of 8 employees.
				Case Study: A company wants to understand the concentration of salaries. How can density plots help?
40	732724322011	E23AI038	Gnanashree V	Theory: What is meant by data standardization?
				Chart: Plot standardized test scores of 6 students.
				Case Study: A school wants to compare scores of two exams with different totals. How can standardization help?
41	732724322012	E23AI036	Fathima Jinan KV	Theory: Explain inequality in the context of income distribution.
				Chart: Draw a bar plot of income of 5 families.
				Case Study: A researcher wants to show income differences among families. How can inequality be visualized?
42	732724322013	E23AI011	Idhaya K	Theory: What is smoothing in time series data?
				Chart: Draw a line plot of daily stock values and apply smoothing.
				Case Study: An investor wants to reduce noise in stock price data. How can smoothing help?
43	732724322014	E23AI008	Karishni M	Theory: Why do we use pivot tables in EDA?
				Chart: Create a pivot table showing monthly sales of two products.
				Case Study: A shopkeeper wants to see monthly sales comparison. How can pivot tables help?
44	732724322015	E23AI022	Mathumitha V	Theory: What is the role of grouping datasets?
				Chart: Draw a bar chart showing average marks of students grouped by gender.
				Case Study: A teacher wants to compare average marks of boys and girls. How can grouping help?
45	732724322016	E23AI003	Mithula S	Theory: Explain the concept of data aggregation.
				Chart: Create a bar chart showing total sales for each quarter.
				Case Study: A company wants to analyze quarterly performance. How can aggregation help?
46	732724322017	E23AI042	Monikajasmine V	Theory: What is the use of Seaborn in data visualization?
				Chart: Create a Seaborn line plot for marks of 5 students.

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				Case Study: A teacher wants more attractive plots than Matplotlib. How can Seaborn help?
47	732724322018	E23AI032	Mounika N	Theory: Define geographic visualization using Basemap.
				Chart: Draw a map showing 3 tourist places in India.
				Case Study: A travel guide wants to show key destinations on a map. How can Basemap help?
48	732724322019	E23AI073	Pachiyammal D	Theory: What are single-variable distributions?
				Chart: Draw a histogram of heights of 10 students.
				Case Study: A sports coach wants to know the height distribution of players. How can this help?
49	732724322020	E23AI049	Poornisha V	Theory: What are numerical summaries of spread?
				Chart: Draw a boxplot showing the spread of exam scores.
				Case Study: A teacher wants to know the range and median of marks. How can boxplots help?
50	732724322021	E23AI071	Priyadharshini M	Theory: Explain scaling in univariate analysis.
				Chart: Create a line plot showing scaled values of student marks.
				Case Study: A tutor wants to normalize student marks out of 1 for easy comparison. How can scaling help?
51	732724322022	E23AI031	Priyadharshini R	Theory: Why do we compare EDA with classical analysis?
				Chart: Draw a scatter plot of study hours vs marks scored.
				Case Study: A student wants to analyze if studying more really improves marks. How can EDA help?
52	732724322023	E23AI007	Priyanka S	Theory: Define Bayesian analysis in simple words.
				Chart: Create a line chart showing probability estimates over time.
				Case Study: A doctor wants to update the probability of recovery based on new data. How can Bayesian analysis help?
53	732724322024	E23AI035	Ranjini V	Theory: What are visual aids in EDA?
				Chart: Draw a bar chart showing monthly electricity usage.
				Case Study: A household wants to track which month electricity usage is highest. How can visual aids help?
54	732724322025	E23AI034	Sandhiya R	Theory: Why is Matplotlib called a core visualization library?
				Chart: Draw a line plot showing sales growth across 6 months.
				Case Study: A small business wants to visualize sales trends. How can Matplotlib help?
55	732724322026	E23AI072	Santhi S	Theory: What are text labels in Matplotlib?
				Chart: Create a bar chart with text labels above bars showing exact values.
				Case Study: A shopkeeper wants to display exact sales numbers on a chart. How can text labels help?
56	732724322027	E23AI061	Santhiya M	Theory: What is 3D plotting useful for?
				Chart: Create a 3D surface plot for a small dataset.
				Case Study: An engineer wants to model machine performance across 3 parameters. How can 3D plots help?
57	732724322028	E23AI056	Saveetha AK	Theory: What are distributions in statistics?
				Chart: Draw a histogram showing exam marks distribution.
				Case Study: A teacher wants to see if marks are normally distributed. How can distributions help?
58	732724322029	E23AI062	Shivani M	Theory: What are spread measures like range and variance?
				Chart: Create a boxplot for salaries of employees.

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				Case Study: A company wants to know if employee salaries are widely different. How can spread measures help?
59	732724322030	E23AI009	Sivadarshini M	Theory: Why is standardizing data important?
				Chart: Draw a scatter plot of standardized marks of 6 students in two subjects.
				Case Study: A teacher wants to compare subject marks fairly. How can standardization help?
60	732724322031	E23AI067	Subashini S	Theory: What is inequality in data analysis?
				Chart: Plot a bar graph showing wealth of 6 families.
				Case Study: A researcher wants to highlight economic inequality. How can this be shown?
61	732724322032	E23AI069	Subashree J	Theory: Why is smoothing applied in time series?
				Chart: Draw a line graph of daily temperatures smoothed over a week.
				Case Study: A weather scientist wants to remove daily noise in temperature records. How can smoothing help?
62	732724322033	E23AI010	Vinothini S	Theory: What is the role of software tools in EDA?
				Chart: Create a bar chart showing attendance in 4 classes.
				Case Study: A teacher wants to summarize class attendance easily. How can tools like Python help?
63	732724322301	E24AILE02	Sravani T	Theory: Why is visualization important in EDA?
				Chart: Draw a scatter plot showing weight vs height of 8 students.
				Case Study: A doctor wants to see the relationship between height and weight. How can visualization help?
64	732724322302	E24AILE03	Varshini	Theory: What is the final goal of exploratory data analysis?
				Chart: Plot a line graph of monthly sales for 6 months.
				Case Study: A shopkeeper wants to decide which months had best sales. How can EDA help in making decisions?